App Rating Prediction using Machine Learning

Project Overview

The purpose of this project it to predict app rating using machine learning techniques. The focus was on improving model performance through feature engineering and data preprocessing.

Objectives

- Predict app ratings based on available features
- Handle categorical and numerical data
- Improve model performance using feature engineering
- Evaluate model accuracy using R^2 score

Dataset

The dataset contains information about mobile apps, including:

- App
- Category
- Content Rating
- Genres
- Reviews
- Installs
- Price
- Rating

Data Preprocessing

Handling Categorical Variables

- Used One-Hot Encoding (pd.get_dummies())

Feature Engineering

- Created:
    - Installs_log using log transformation
    - Reviews_logusing log transformation
- This helped reduce skewness and improve model performance

Model Used

- Linear Regression

Model Performance

Metric	Score
Training R^2	0.54
Test R^2	0.58

Key Insights

- Log transformation of installs significantly improved model performance
- Simpler models with strong features performed better than complex models
- Higher installs generally lead to higher ratings

Conclusion

The model achieved good performance with an R^2 score of 0.58.
Feature engineering played a key role in improving model accuracy.

Tools & Technologies

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Future Improvements

- Try advanced models like Random Forest or XGBoost as the features available do not influence the app rating linearly
- Hyperparameter tuning

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